

Assignment M4

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Abstract—Over the course of the M assignments, I will be investigating the task of searching for local events, with a focus on the Meetup.com website. Meetup.com’s search page currently has two main page views and several filters, but there are some use cases that are difficult to accomplish (for example, searching for certain types of events).

1 QUALITATIVE EVALUATION

1.1 Selection

This evaluation will be for a paper prototype and the evaluation method will be a survey. *Appendix 5.1: Paper Prototype for Qualitative Evaluation (from M3)* shows a picture of the prototype.

1.2 Evaluation Plan

The survey will be hosted on [Peer Survey](#) so that participants can take the survey asynchronously. Participants will primarily be classmates in this class, CS6750. If I find that it is difficult to reach the desired number of participants (25), then I may also ask friends and family to take the survey. Classmates will be recruited through the participant recruitment threads on [Ed](#), and friends and family will be individually texted if I need their additional help. It will not be recorded, only the data from the survey will be analyzed.

1.3 Content of the Evaluation

First, I will ask some demographic questions of the user (age, what type of area they live in, etc.) so that I can see if that affects their answers. Then, I will ask them specific questions about the prototype, as listed below in Table 1.

1.4 Evaluation of Requirements and Data Inventory

One of the requirements from M2 was that novice users should be able to see the filter(s) that they want from the home page and should be able to use that filter in only 1-2 clicks. Another requirement was that users can filter by genre, date, and location, which this prototype allows for. Both of these requirements were

Table 1—The questions for the survey

Question	Answer Type and Options
Select your age:	Age Group (Under 18, 18-29, 30-39, 40-49, 50-64, 65+)
Which of the following best describes the area you live in?	Choose one (Urban, Suburban, Rural)
Please think back to a recent year that was unaffected by COVID-19 (e.g. 2019). Approximately how many times in that year did you search for a local meetup?	Choose one (0 times per year, 1-3 times per year, 4-6 times per year, 7-10 times per year, 11+ times per year)
Please think back to a recent year that was unaffected by COVID-19 (e.g. 2019). Approximately how many times in that year did you attend a local meetup?	Choose one (0 times per year, 1-3 times per year, 4-6 times per year, 7-10 times per year, 11+ times per year)
When searching for a meetup, what is your typical goal?	Short answer
How often do you think you would use the Genre filter when searching for a meetup?	Frequency (Very Frequently, Frequently, Occasionally, Rarely, Never)
How often do you think you would use the Location filter when searching for a meetup?	Frequency (Very Frequently, Frequently, Occasionally, Rarely, Never)
How often do you think you would use the Date(s) filter when searching for a meetup?	Frequency (Very Frequently, Frequently, Occasionally, Rarely, Never)
How often do you think you would use the Time(s) filter when searching for a meetup?	Frequency (Very Frequently, Frequently, Occasionally, Rarely, Never)
How often do you think you would use the Number of Attendees filter when searching for a meetup?	Frequency (Very Frequently, Frequently, Occasionally, Rarely, Never)
Are there any filters you would recommend adding to this interface?	Short answer
This interface looks easy to use.	Agreement (Strongly Agree, Agree, Neutral, Disagree, Strongly Disagree)
This interface looks enjoyable to use.	Agreement (Strongly Agree, Agree, Neutral, Disagree, Strongly Disagree)
Assuming that a meetup relevant to my interests exists, I am confident that I can find it by using this interface.	Agreement (Strongly Agree, Agree, Neutral, Disagree, Strongly Disagree)
The way the results are displayed in the interface is easy to understand.	Agreement (Strongly Agree, Agree, Neutral, Disagree, Strongly Disagree)
What do you like about this interface?	Short answer
What do you not like about this interface?	Short answer
Do you have any suggestions or other comments about this interface?	Short answer

based on the data inventory sections of the users' tasks and subtasks to filter by certain features. These survey questions ask participants how useful each filter is to them and if there is any filter missing from the list. That way, we can see if the prototype fulfills this need.

One aspect of the data inventory was that users' goals were to find fun events such as social and recreational/sports meetups. The survey also asks users what their goal is for using the interface, so we can see if the participants of this survey have the same goal and if that goal is met by the prototype.

The data inventory from M2 also found that users were novices at finding events and were located primarily in urban or suburban communities, so the questions in this survey confirm that these participants are also novices in those regions, to make sure that their needs are being met.

2 EMPIRICAL EVALUATION

2.1 Selection

This evaluation will be for a verbal prototype. *Appendix 5.2: Verbal Prototype for Empirical Evaluation (from M3)* shows the text of the prototype.

2.2 Control and Experimental Conditions

For this evaluation, I will have participants test out the prototype and compare those results with those from the real Meetup.com website. Therefore, the control condition, or independent variable, is the interface - the prototype versus the Meetup.com website. This assumes that the prototype and Meetup.com website have the same dataset of events and the filtering capability is accurate (so if the user clicks to filter by genre, the events will filter appropriately).

In this experiment, I will have participants walk through a number of scenarios to find a goal (e.g. find a running event in the next two weeks). The experimental conditions, or dependent variables, are how many steps or clicks it takes to accomplish that goal (find a relevant event) on both the prototype and the Meetup.com website, and how many of the first 10 results are relevant for that goal. The different tasks/goals are described in the Experimental Method section below.

2.3 Null and Alternative Hypotheses

The null hypothesis is that there is no difference in the number of steps the user has to take to accomplish each goal on the prototype and on the Meetup.com website, and the number of relevant results in the first 10 results is the same for both treatments.

The alternative hypothesis is that there is a statistically significant difference in the number of steps the user has to take to accomplish each goal on the prototype and on the Meetup.com website, and the number of relevant results in the first 10 results is statistically significantly different for both treatments.

2.4 Experimental Method

This will be a within-subjects experiment. Participants will be randomly assigned to each group of which treatment they get first (prototype or Meetup.com).

The different tasks/goals that participants will try to accomplish for each treatment are:

- Finding a particular genre of event (for example, a running meetup)
- Finding events over a range of dates (for example, this weekend)
- Finding the next time a particular group is meeting up (for example, a specific book club)
- Combining multiple filters (for example, finding a running meetup next weekend)

For each of those situations, the data generated will be how many clicks or steps does the participant need to take in order to accomplish that goal. For the verbal prototype this might be a bit unusual, because the prototype involves the participant filling out a questionnaire before seeing results. Therefore, the questionnaire might be too many steps for simple situations, and this evaluation will reveal that.

For both treatments, first the person conducting the evaluation will describe the goal (e.g. "find a running meetup"). Then, for the prototype treatment, the participant will fill out the entire questionnaire with that goal in mind. At the end of the questionnaire, the number of steps that the participant took (number of questions they had to answer, buttons they had to press) will be one data point. The other data point will be the number of events in the top ten results

that are relevant to the goal. For the Meetup.com treatment, the participant will use different filters and search functionality on the website to fulfill the same goal. When the participant is done, the number of clicks/steps taken will also be recorded, as will the number of relevant results in the first ten results.

When I have data from all of these evaluations, because the number of clicks and number of relevant results are both ratio data, I can use a Student's t-test for analysis on the data to see if the null hypothesis or the alternative hypothesis is correct.

2.4.1 Lurking Variables

One lurking variable is which treatment (prototype or Meetup.com) the participant tries first. They might remember from the first treatment details about the events that would make their search in the second instance easier or they might change their answers to the questionnaire (for the verbal prototype) to find the right events. For example, if one of the goals is to find a running meetup, and the participant remembers that they found a running meetup on July 25th, they might filter by July 25th and find it faster even though that's not how they would normally search for a running event.

Another lurking variable is they might already be familiar with the Meetup.com website. This prior knowledge might make them better at using the site and therefore they might be more efficient as searching and filtering for events.

3 PREDICTIVE EVALUATION

3.1 Selection

This evaluation will be for a textual prototype and the evaluation method will be performing a cognitive walkthrough. *Appendix 5.3: Textual Prototype for Predictive Evaluation (from M3)* shows the text of the prototype.

3.2 Tasks for the Cognitive Walkthrough

For the cognitive walkthrough, I will be focusing on how novice users would approach the interface. The data inventory from M2 showed that users do not search for meetups often, so this prototype needs to have high learnability. I will then try out the same tasks that I did in the needfinding exercises:

- Finding a particular genre of event (for example, a running meetup)

- Finding events over a range of dates (for example, this weekend)
- Finding the next time a particular group is meeting up (for example, a specific book club)
- Combining multiple filters (for example, finding a running meetup next weekend)

For each of those tasks, the user will need to first enter in their location in the search bar at the top of the page so the map will recenter on their location (city or zip code). Then, the user can use the filter at the top to search for a particular date range. The icons that show up on the map will tell the user what type of event and where it is located, then when they tap each icon it will give them more information about the event. I will be evaluating how easy it is for a user to navigate around the interface so that they can accomplish each of these goals.

Ideally, the interface allows for users to play around with the map view and try out different filters, locations, icons, etc. without fear of making mistakes. Part of what I will evaluate is how easy it is to "undo" a decision if the user makes a mistake or filters too much.

The cognitive walk-through will also help evaluate if there needs to be more information on the main screen, such as descriptions of the icons, more filters, etc. and what information is most useful in the "detail" view of each event in order to accomplish the goals listed above.

4 PREPARING TO EXECUTE

For the next assignment, I will complete the qualitative and predictive evaluations.

The qualitative evaluation will be relatively easy to conduct because the questions are already written out and it is simple to use the Peer Survey tool to create the survey. Also, many students are participating in each others' surveys, so recruiting many participants should be relatively simple. It also seems that this survey will gather a lot of useful information, which I am interested to see.

The predictive evaluation will also be relatively easy to conduct because I can do it with a low fidelity prototype and no extra participants or equipment. The tasks that I have laid out are tasks that I have tried during the needfinding portions on existing websites such as Meetup.com, so it will be interesting to

try them again on the prototype and see how it addresses the concerns laid out earlier in the assignments.

Finally, I decided not to choose the empirical evaluation because my prototype is not fleshed out enough to show accurate information (such as search results) and it would be difficult to recruit people for such a long and detailed interview process.

5 APPENDICES

5.1 Paper Prototype for Qualitative Evaluation (from M3)

This is shown in Figure 1, below.

5.2 Verbal Prototype for Empirical Evaluation (from M3)

For this prototype, I would ask users questions based on the questionnaire that would actually be in the app. So for example, I would first ask users what their main goal is for finding an event. If they said they were trying to meet people for example, then that would prioritize events that are either recurring (i.e. a weekly run club) or have a large attendance. On the other hand, if their main goal was to just get out of the house, then a one-off event with few attendees might accomplish their goals. If their goal was to learn a certain skill or get in shape, the app would then only show events that related to that activity or exercise. Their goal might also determine what time range the events should be in; if they want to get out of the house this weekend, then that would automatically limit the events to that date range.

If their goal did not already limit their activity, then I would ask users about what they enjoy doing; this could be any activity they have enjoyed in the past or think they might enjoy, such as running, crocheting, reading, etc. This would help show a diverse range of events that a user might be interested in. If their goal already answered this question, then the question would be skipped.

Then I would ask users about their availability and date range. This could either be their general availability (such as "weeknights after 5 PM") or more specific (such as "this Tuesday between 7 and 9 PM"). The date range might also be more immediate (an event today) or broader (this month). Again, this would limit the events that would be shown in the app.

Meetup.com

FILTER BY Show/Hide

Genre

☒ Running
☒ Book Club
☒ Soccer

Location

☒ 78704
☒ 78705

From 7/4/21 To 7/6/21

Date(s)

From 5:00 PM To 8:00 PM

Time(s)

At Least 5

Number of Attendees

FILTER

RESULTS

ATX Women Run 7/5/21 at 6 PM in 78704	Running	11 Attending
Historical Fiction Book Club 7/6/21 at 7 PM in 78705	Book Club	6 Attending
Science Fiction Book Club 7/5/21 at 8 PM in 78704	Book Club	22 Attending
Co-Ed Soccer League 7/4/21 at 5 PM in 78704	Soccer	74 Attending
Amstn Soccer Club 7/4/21 at 7 PM in 78705	Soccer	32 Attending

Figure 1—A paper prototype of the filters and results page

Then I would ask users about what locations they would travel to. This could be a wide range of answers. For example, perhaps they have a long commute to work, so on weeknights they want to stay near their work but on the weekends they want to stay near their home. This again would filter down the events.

At the end of the questionnaire, the app would show a list of the events sorted by most relevant to their goals, availability/dates, locations, and activities. As they scrolled down, the events shown are less relevant, but they could keep scrolling until they found the last event that matched at least one of these attributes. Each event would also show what attributes were matched (why it was deemed relevant) so that the user could quickly understand its relative ranking, and which attributes were not matched so that users could understand the trade-offs (for example, maybe it fulfills their goal and location but is earlier in the day than they specified).

5.3 Textual Prototype for Predictive Evaluation (from M3)

For this idea, the main app view (both for mobile and web - smartphones, laptops, etc.) would be a map. If the user allows sharing their location with the app, then the map would be centered around their current location with a blue circle indicating where they currently are. If the user does not allow sharing their location, the map would be defaulted to a particular area (for example, many apps default to San Francisco) and then users can search in a search bar at the top of the screen for a particular city or zip code.

Once users have navigated to the correct location in the map, then they can start using their fingers to drag and zoom in/out of the map. There will be an icon at each location on the map that has an event coming up in the next two weeks by default. There is a date picker at the top of the screen so that users can adjust this date range to be whatever they want (to show only events happening today, for example). The icon will correspond to the general genre of the event (a book icon for book club, for example).

If the user clicks on icon, a summary view pops up showing the event's name, location, date/time, and a brief description. If they click the X on the summary view popup, it closes the view. If they click the "More" button on the summary view popup, it navigates to a new page with more information (this new page will have a back button so users can return to the map easily). If the same location has more than one event in the next two weeks, then the summary view

would only show event titles and date/times and users would have to click the "More" button to see the descriptions.

Users can favorite certain venues or certain locations (like their home or work) so that those locations are highlighted with a star on the map for future reference. This view does not currently allow any filters other than location and date so that the map is the main focus, though that may change in the future.